

Homework 3 – Generative AI (CS/DS 552, Whitehill, Fall 2026)

Collaboration policy: You may complete this assignment with a partner, if you choose. In that case, both partners should sign up on Canvas in a **pre-made group** as a team, and only one of you should submit the assignment. You are permitted to use ChatGPT (or another AI tool) to help with (a) data visualization and (b) general PyTorch and Python syntax. You may *not* use it to generate code that is specific to this assignment (i.e., anything about GANs).

1. Generative Adversarial Networks (GANs) for Face Generation [25 pts]

(a) Unconditional generation [20 pts]:

Using the same dataset as in Homework 2 part 1, train a GAN to generate face images. Show a collage of 20 generated faces after training. Report any patterns you observe, e.g., do you see evidence of mode collapse?

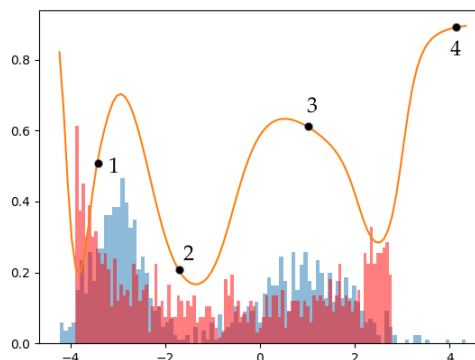
(b) Conditional generation [5 pts]:

Now, using the data in `ages23k.npy` (containing the ages of the face images from the previous part), train a **conditional** GAN so that, given both a random noise vector $\mathbf{z}$ and an additional value y , the generator will generate a face image $\mathbf{x}$ with a target age y . Show a 6x10 collage, where row $i \in \{1, \dots, 6\}$ contains 10 faces of people generated with a target age $y = 8 + i * 10$. **Note:** the age labels for this face dataset were obtained from Amazon Mechanical Turk and are thus not completely accurate. Do not be surprised if you find a few mislabeled data.

2. Generative Adversarial Networks (GANs) for 1-d Data [20 pts]

(a) Explain D 's Outputs [4 pts]:

Suppose we train a GAN to generate 1-d data to match a given data distribution $P_{\text{data}}(x)$. The figure below shows two histograms (red and blue) as well as an orange curve. The blue histogram is the true data distribution $P_{\text{data}}(x)$. The red histogram is the generated data distribution $P(G(z))$ of a trained GAN generator G whose input $z \sim \mathcal{U}(0, 1)$. The orange curve shows the output $D(G(z))$ of the trained discriminator D .



For each of the four labeled points 1, 2, 3, 4 in the figure, explain in qualitative terms why the corresponding outputs of the discriminator $D(x_1), D(x_2), D(x_3), D(x_4)$ equal their respective values 0.5, 0.2, 0.6, and 0.9.

(b) Show How a GAN Learns to Approximate $P_{\text{data}}(x)$ [16 pts]

Similar in spirit to Homework 2, part 2, this is an open-ended exercise: Choose some true data distribution $P_{\text{data}}(x)$ (where, for simplicity and ease of visualization, $x \in \mathbb{R}$), choose a GAN architecture, and show how, over the course of GAN training, the generator can learn to approximate

$P_{\text{data}}(x)$ more and more closely. In addition to your code, you should submit a sequence of 10 graphs (one for every few epochs of training) containing the same 2 histograms and orange curve as in the previous exercises, but tailored to your $P_{\text{data}}(x)$, $P(G(z))$, and D .

Submission: Create a Zip file containing all your Python code (which can be in multiple files) and your PDF file, and then submit on Canvas. If you are working as part of a group, then only **one** member of your group should submit (but make sure you have already signed up in a pre-allocated team for the homework on Canvas).